Journal of Neuroscience Methods 260 (2016) 83–90

Contents lists available at ScienceDirect

Journal of Neuroscience Methods

journal homepage: www.elsevier.com/locate/jneumeth

The *in vitro* isolated whole guinea pig brain as a model to study epileptiform activity patterns

Marco de Curtis*, Laura Librizzi, Laura Uva

Unit of Epileptology and Experimental Neurophysiology, Fondazione Istituto Neurologico Carlo Besta, Milano, Italy

HIGHLIGHTS

- The *in vitro* isolated brain retains the connectivity between remote brain areas.
- Acute epileptiform activities can be induced by simple pharmacological manipulations.
- Epileptiform patterns determinants can be identified in a close-to-*in vivo* condition.

ARTICLE INFO

Article history:
Received 30 December 2014
Received in revised form 23 March 2015
Accepted 24 March 2015
Available online 2 April 2015

Keywords:
Ictogenesis
Interictal
In vitro guinea pig brain
Seizures

ABSTRACT

Background: Research on ictogenesis is based on the study of activity between seizures and during seizures in animal models of epilepsy (chronic condition) or in *in vitro* slices obtained from naïve non-epileptic brains after treatment with pro-convulsive drugs, manipulations of the extracellular medium and specific stimulation protocols.
New method: The *in vitro* isolated guinea pig brain retains the functional connectivity between brain structures and maintains interactions between neuronal, glial and vascular compartments. It is a close-to-*in vivo* preparation that offers experimental advantages not achieved with the use of other experimental models. Neurophysiological and imaging techniques can be utilized in this preparation to study brain activity during and between seizures induced by pharmacological or functional manipulations.
Results: Cellular and network determinants of interictal and ictal discharges that reproduce abnormal patterns observed in human focal epilepsies and the associated changes in extracellular ion and blood–brain permeability can be identified and analyzed in the isolated guinea pig brain.
Comparison with existing methods: Ictal and interictal patterns recorded in *in vitro* slices may show substantial differences from seizure activity recorded *in vivo* due to slicing procedure itself. The isolated guinea pig brain maintained *in vitro* by arterial perfusion combines the typical facilitated access of *in vitro* preparations, that are difficult to approach during *in vivo* experiments, with the preservation of larger neuronal networks.
Conclusions: The *in vitro* whole isolated guinea pig brain preparation offers an unique experimental model to study systemic and neurovascular changes during ictogenesis.

Published by Elsevier B.V.

Contents

1. Recording brain activity in the IWB 84
2. Blood–brain barrier, inflammation and adhesion factors during seizures 87
3. Region-specific epileptiform patterns in the IWB 87
4. Conclusions 89
Acknowledgements 89
References 89

* Corresponding author. Tel.: +39 0223942280.
E-mail address: decurtis@istituto-besta.it (M. de Curtis).

http://dx.doi.org/10.1016/j.jneumeth.2015.03.026
0165-0270/Published by Elsevier B.V.

Epilepsy is a comorbidity of diseases due to diverse pathogenic factors that result in paroxysmal changes of brain activity. Both systemic or focal alterations may contribute to set a chronic epileptic condition characterized by abnormal excitability that underlies the spontaneous generation of seizures. Several pathogenic factors (genetic, metabolic, degenerative, *etc.*) activate an epileptogenic process that may involve different neuronal, glial and neurovascular brain compartments. The processes that lead to a chronic epileptic condition are the target of research on *epileptogenesis*. The analysis of the mechanisms that generate a seizure is the study of *ictogenesis*.

Research on ictogenesis is based on the study of activity between seizures and during seizures in animal models of epilepsy (chronic condition) and seizures. In the latter group of models, interictal and seizure discharges are studied in normal non-epileptic brains following acute pharmacological and electrical manipulations that promote seizures. *In vitro* brain slice preparations have been largely utilized since early 1980s to study ictogenesis. Seizures are most frequently analyzed on *in vitro* slices obtained from naïve non-epileptic brains after treatment with pro-convulsive drugs, manipulations of the extracellular medium and specific stimulation protocols (see Pitkänen et al., 2006). Similar treatments are also utilized to induce seizure activity on *in vitro* slices from epileptic animals, since brain slices do not generate spontaneous seizure activity, but may produce interictal events such as spikes or other interictal patterns.

Ictal events and interictal patterns recorded in slices *in vitro* may show substantial differences from seizure activity recorded *in vivo* (see de Curtis and Avanzini, 2001; de Curtis and Gnatkovsky, 2009). These differences are likely due to the reduction of local networks preserved within a slice and to the alteration of the neuro-glio-vascular interactions associated with the slicing procedure. To overcome slice tissue limitations, more complex *in vitro* models have been developed, that combine the typical facilitated access of *in vitro* preparations with the preservation of larger neuronal networks. The isolated guinea pig brain maintained *in vitro* by arterial perfusion represents one of these preparations (de Curtis et al., 1991, 1998a, 2012; Muhlethaler et al., 1993). In the *in vitro* isolated whole brain (IWB) of the guinea pig, the intrinsic connectivity within and between diencephalic and telencephalic structures of both hemispheres is maintained (Uva and de Curtis, 2005). Moreover, the vascular system, the blood–brain barrier and their interaction with the brain parenchyma is functionally preserved (de Curtis et al., 1998a; Librizzi et al., 2001, 2012). The study of ictogenesis in the IWB has the same limitation of other *in vitro* preparation with respect to the ability to generate spontaneous seizures. Pro-epileptic drugs and high-frequency stimulations must be employed, indeed, to obtain seizure activity in the IWB. Seizure patterns in the IWB closely resemble seizures recorded *in vivo* in rodents models of epilepsy and in humans (see below). In the last 20 years of experimental work on the guinea pig IWB, we mainly focused on the characterization of system physiology of the olfactory and limbic systems. In these regions, we reproduced interictal and seizure patterns identified in human epilepsies to study cellular activity, network interactions, extracellular ion changes and neurovascular alterations responsible for ictogenesis.

1. Recording brain activity in the IWB

The most frequently explored areas in the IWB are the olfactory and limbic regions: activity have been recorded in the olfactory bulb (OB; Uva et al., 2006; Carriero et al., 2009), in the piriform cortex (PC; Biella and de Curtis, 1995, 2000; Biella et al., 1996), in the amygdala (Carriero et al., 2009), in the entorhinal and perirhinal cortex (EC and PRC; Biella et al., 2000, 2002; Gnatkovsky et al., 2004) and in the hippocampal region such as the CA1 region, the dentate gyrus (DG) and the subiculum (Boido et al., 2014; Uva and de Curtis, 2003, 2005). These regions can be easily reached in the IWB when the brain is positioned in the recording chamber with its ventral surface exposed (Fig. 1). This represents an advantage compared to the *in vivo* condition, in which cortical structures positioned ventrally at the base of the skull are difficult to access. In the IWB, the position of recording and stimulating electrodes is defined under direct visual control with a stereomicroscope and can be rapidly modified during the experiment. For deep structures, evoked field potentials are utilized to drive the electrodes to the targets. This offers a virtually unlimited accessibility to stimulate and record the ventral brain and the related depth nuclei/structures. The viability of recorded areas can be evaluated in the IWB by monitoring evoked potentials along multi-synaptic pathways following electrical stimulation of the lateral olfactory tract (Fig. 1) or any other brain area. Another advantage of the IWB compared to *in vivo* conditions, is the mechanical stability of the recordings, granted by the absence of respiration and pulsation artifacts (see de Curtis et al., 1991, 1998a; Muhlethaler et al., 1993).

The possibility to perform simultaneous field potential (FP) recordings from different regions on both hemispheres makes the IWB the ideal preparation to study propagation of both normal and epileptiform activity in anatomically connected brain regions. FPs are recorded with different types of extracellular electrodes: glass micropipettes filled with 0.9% NaCl; tungsten or stainless steel wires with variable tip diameter, and multichannel electrodes with diverse arrangements of recording sites. Multi-site recordings performed in different cortical areas with glass micropipettes during seizure activity induced pharmacologically by arterial perfusion of 50 μM bicuculline methiodide (BMI) are illustrated in Fig. 2. In this acute model of ictogenesis, simultaneous extracellular recordings performed in the PC, in the lateral and medial EC, in different subfields of the hippocampal formation (CA1 and DG) demonstrate that seizure-like events are generated in EC-hippocampus (Librizzi and de Curtis, 2003; Uva et al., 2005). We also demonstrated that, as in chronic models of temporal lobe epilepsy (Bragin et al., 1999) and in human mesial temporal lobe epilepsy (Bragin et al., 2005), in this acute model of temporal lobe seizures two different patterns of seizure initiation, defined as low-voltage fast activity (FA) and hypersynchronous activity (HSA) were observed (Boido et al., 2014). In this study, simultaneous extracellular FP recordings performed from ventral hippocampal and parahippocampal subregions demonstrated that both patterns are locally generated within the limbic area. Medial-EC, CA1 and subiculum are involved in all phases in both FA- and HSA-onset seizures. Moreover, we showed that the DG has an important role in the maintenance and development of the seizure, but is inactive at the onset of FA seizures. Surprisingly, pre-subiculum and para-subiculum provide minimal participation to seizure activity, despite the documented anatomical connections with the EC and the hippocampal formation.

To analyze local activations in a laminated structure (*i.e.*, cortical regions) 16-ch linear silicon probes (50 or 100 μm contact separation; Neuronexus, Ann Arbor, MI, USA) are inserted perpendicular to cortical lamination in the structures of interest (Boido et al., 2014; Uva and de Curtis, 2005; Uva et al., 2005). FPs laminar profiles are recorded with silicon probes and are then analyzed by applying current-source-density (CSD) analysis (de Curtis et al., 1994; Biella and de Curtis, 1995), which is a useful method to distinguish locally generated components from far field components passively propagated in the tissue, and to characterize spatial and temporal arrangement of locally generated responses. Electrolytic lesions performed at the end of the experiments are used to reconstruct the position of the silicon probes tracks on histological sections. The use of multichannel silicon probes to perform laminar

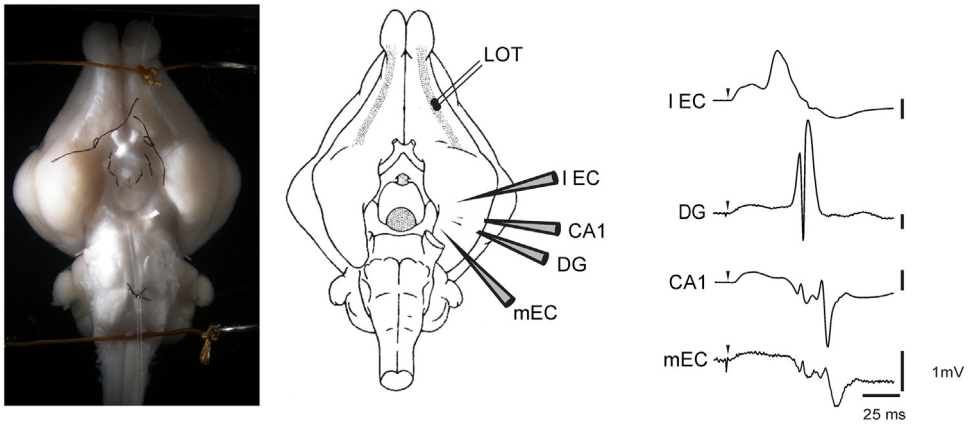

Fig. 1. On the left, a photograph of the isolated guinea pig brain maintained *in vitro*. In the middle, a scheme of the brain is illustrated to show the position of the stimulating electrode placed on the lateral olfactory tract (LOT) and the recording glass pipettes inserted in the lateral and medial entorhinal cortex (EC) and in the dentate gyrus (DG) and CA1 area of the hippocampus. On the right, the electrical stimulation of the LOT evoked potentials simultaneously recorded in the structures shown in the brain drawing.

analysis of epileptiform interictal and ictal discharges has been crucial to demonstrate the local generation of fast activity at the onset of BMI-induced ictal discharges in CA1 and medial entorhinal cortex (Uva et al., 2005) and the subsequent involvement of dentate gyrus in the seizure (Boido et al., 2014).

When the aim of the experiment is to monitor the tangential propagation of the activity across a cortical region, stainless steel microelectrode matrix arrays can be utilized to record simultaneously from different closely spaced positions (Dickson et al., 2003; Gnatkovsky et al., 2004). An example of seizure recorded with a 16 channel array (4 × 4 recording sites separated by 500 μm; FHC, Bowdoinham, USA) in the medial EC of the IWB after 50 μM BMI arterial perfusion is illustrated in Fig. 3.

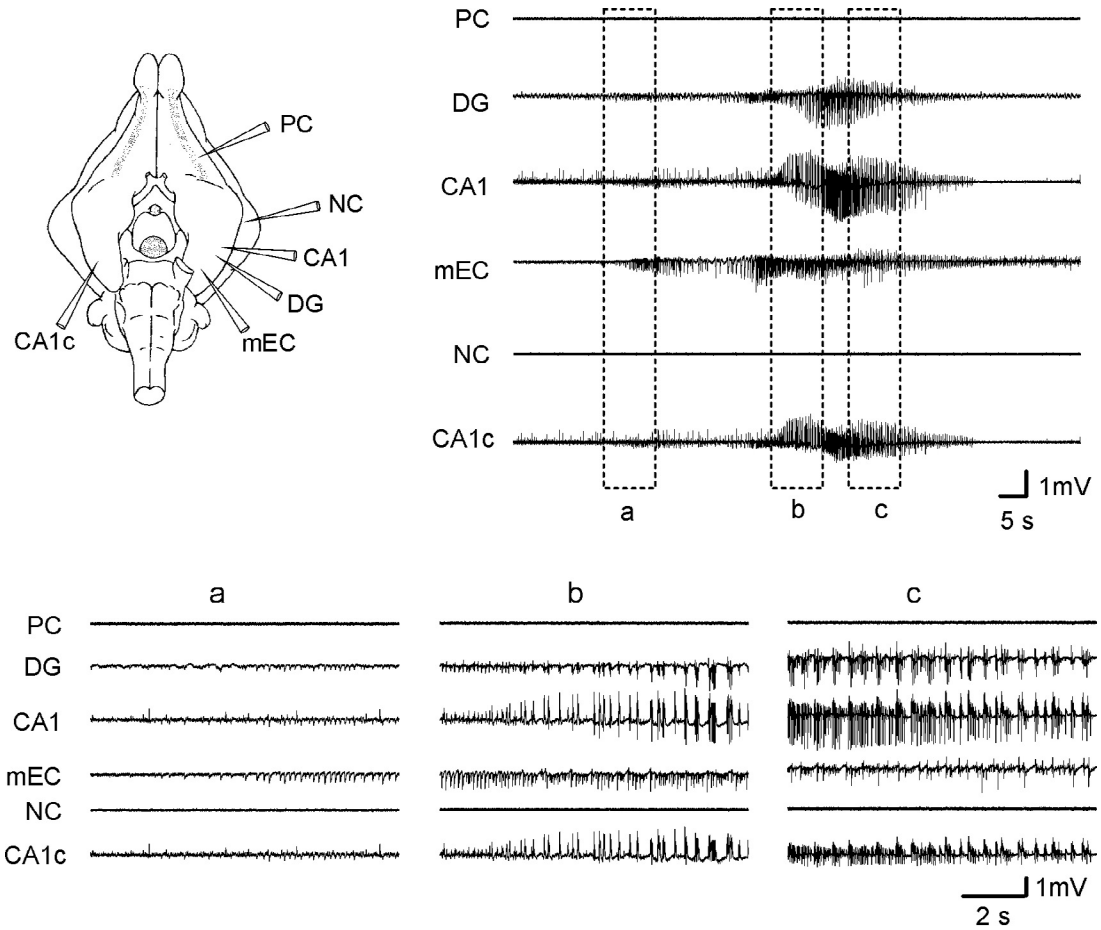

Fig. 2. On the top left, a scheme of the isolated guinea pig brain showing the position of the recording glass pipettes inserted in piriform cortex (PC), neocortex (NC), medial entorhinal cortex (mEC), dentate gyrus (DG) and hippocampus (CA1 bilaterally). On the right, representative extracellular recordings performed from the same areas, following systemic perfusion of KA (6 μM; 50 ml). Expanded EEG segments of interest (dashed boxes) are reported in the lower panels.

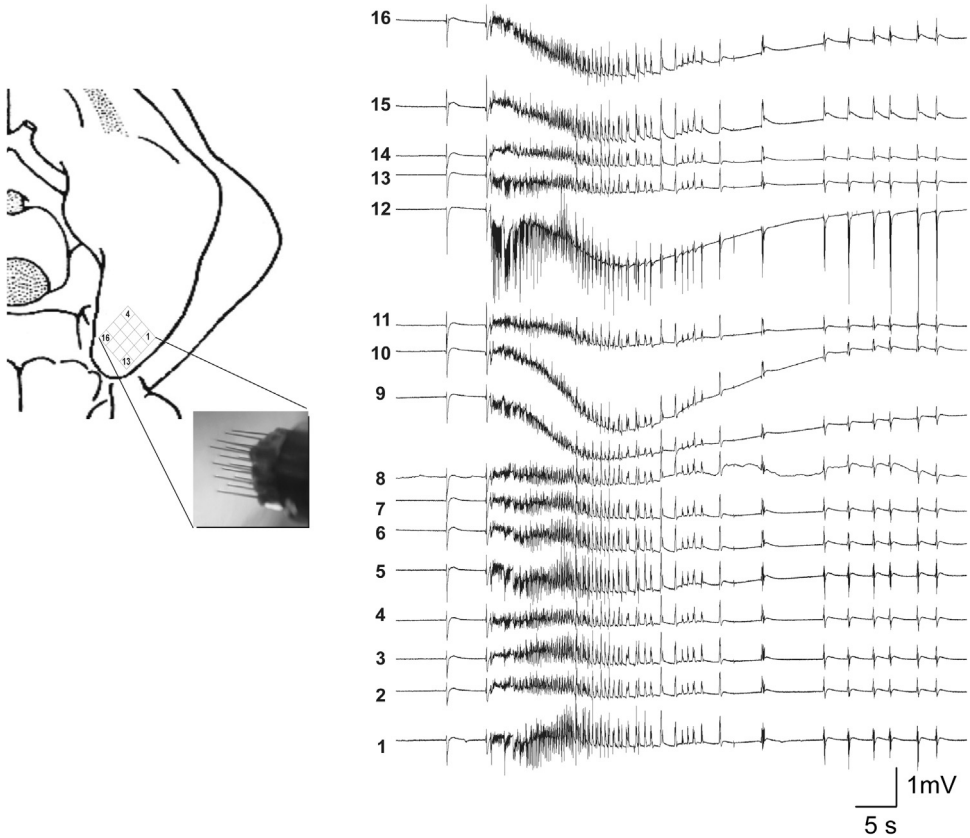

Fig. 3. On the left, a scheme of a portion of the isolated guinea-pig brain showing the position of 4 × 4 stainless steel electrodes matrix array recording in the medial entorhinal cortex (mEC). A microphotograph of the electrodes inserted in the tissue is shown in the inset. On the right, seizure-like events induced by arterial application of the $GABA_A$ receptor antagonist, bicuculline methiodide (50 μM, 3-min) are simultaneously recorded in 16 different positions in mEC with the 16-channel matrix electrode.

The study of the propagation of neuronal activity can be achieved in the IWB by implementing optical imaging of signals generated by voltage-sensitive dyes (VSD). Fast VSD-dependent imaging of stimulus-evoked activity in conditions of normal excitability was performed in the olfactory and limbic cortices following arterial perfusion of different types of VSD (RH795 in de Curtis et al., 1999b; 4- and 8-ANEPS in Biella et al., 2003 and Gnatkovsky and de Curtis, 2006). Seizure-like activity was also recorded in the entorhinal–perirhinal areas after local applications of BMI in cortical areas 35–36 (Biella et al., 2010). Intrinsic optical signals generated by interictal events in the IWB have been also analyzed (Federico et al., 1994).

FP recordings can be coupled with intracellular recordings from principal cells and interneurons, performed with sharp micropipettes filled with 2 M potassium acetate and 1.5% biocytin (60–120 MΩ input resistance). The position of the intracellular electrodes is visually controlled and its depth is defined by the micrometric scale of the micromanipulator utilized for advancing the recording electrode during the experiment. The morphology of the recorded cell is verified histologically at the end of the electrophysiological experiment by revealing HRP conjugated with biocytin injected at the end of the intracellular recording with visualization (de Curtis et al., 1999a; Forti et al., 1997; Pare et al., 1992; Gnatkovsky et al., 2007, 2008). The combination of simultaneous intra and extracellular recordings is useful to define the intrinsic neuronal properties of single cells and to study the pattern of neuronal activity during the generation of extracellular potentials and therefore to identify the nature of the extracellular components observed in physiological conditions and during epileptiform activity (Biella et al., 2002; Gnatkovsky et al., 2008; Uva et al., 2009, 2015). Simultaneous extracellular and intracellular recordings in the IWB medial EC were useful to demonstrate that both the preictal spikes induced by arterially perfusion of BMI or 4-AP (Uva et al., 2015) and the fast 20–30 Hz activity recorded at the onset of a focal seizure are mediated by inhibitory networks (Gnatkovsky et al., 2008).

Neuronal activity is associated with functionally significant changes in both extracellular and intracellular concentrations of pH and other ions, such as K^+ and Ca^{2+}, that can be monitored with ion-sensitive microelectrodes. Double-barreled glass pipettes recordings were utilized in the IWB to simultaneously record at the same site FPs, extracellular K^+ and protons (H^+ as a measure of extracellular pH) signals with ion-selective resins (de Curtis et al., 1998b; Librizzi et al., 2001; Gnatkovsky et al., 2008; Uva et al., 2008). Local application of BMI in the second layer of the PC was able to induce local interical spikes that spread from the PC to the lateral EC and to the hippocampal formation. Spontaneous interictal spikes recorded extracellularly in the PC consistently correlated with a rapid increase in extracellular K^+ concentration by 0.5–1 mM. Similarly, a fast-rising extracellular alkaline shift (ranging between 0.05 and 0.08 pH units) was observed during the inter-spike period (de Curtis et al., 1998b). The recovery to control values coincided with the activation of the next interictal spike, suggesting that the measured ionic changes are involved in pacing interictal spike periodicity at 0.1–0.2 Hz. Arterial application of BMI induced focal epileptiform activity in limbic areas, initiated by low-voltage fast activity correlated with interneurons firing and IPSPs in principal neurons. This seizure onset pattern was associated with a rapid rise in extracellular K^+ (Gnatkovsky et al., 2008; Trombin et al., 2011; Fig. 4) and with biphasic changes in extracellular pH characterized by an early low-amplitude alkalinization (asterisks in Fig. 4) followed by a prominent and long-lasting acidification.

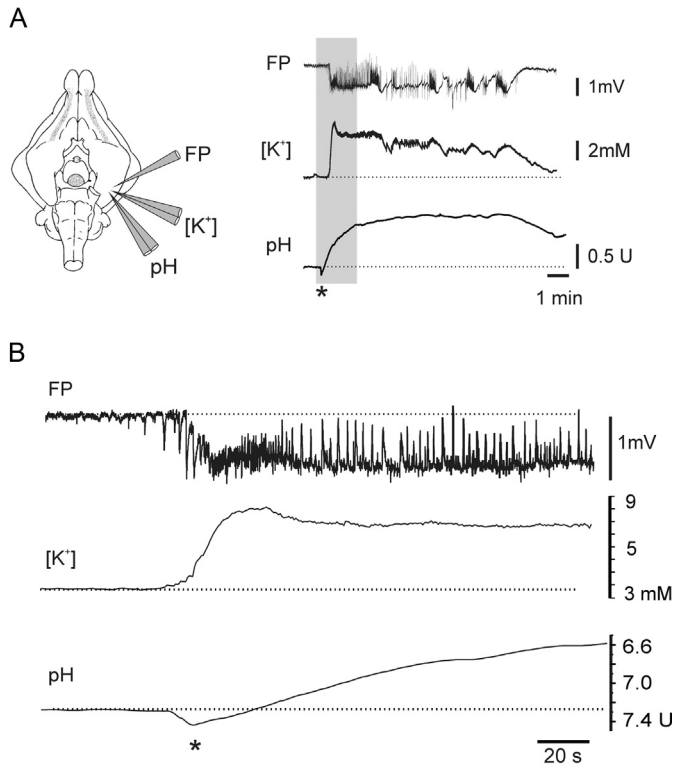

Fig. 4. On the left in A, scheme of the isolated guinea-pig brain showing the position of the double barreled ion-sensitive electrodes (K^+ and pH) inserted in medial entorhinal cortex (mEC). A recording glass pipette is also inserted in the same area. Arterial application of bicuculline methiodide (50 μM; 3 min), induced seizure like activity associated with changes in extracellular potassium concentration and pH levels. In B, the area in A shaded in gray is expanded.

2. Blood–brain barrier, inflammation and adhesion factors during seizures

The blood–brain barrier (BBB) represents the highly regulated interface between the brain parenchyma and the plasma compartment that contributes to maintain the homeostasis of the cerebral microenvironment. BBB function has been found to be primarily altered in several neurological disorders and in epilepsy (Cornford, 1999; de Vries et al., 1997; Marchi et al., 2007). Many attempts have been made to reproduce morphological and physiological BBB properties in a facilitated experimental condition. The IWB maintained *in vitro* by arterial perfusion represents an unique model in which the complex interactions between the vascular and the neuronal compartments are preserved (de Curtis et al., 1998a; Librizzi et al., 2001; Mazzetti et al., 2004). Electron microscopy after 5 h *in vitro* demonstrated that cerebral capillaries and BBB specializations in IWB retain features compatible with structural integrity. Following arterial application of compounds acting on brain excitability in the IWB (*i.e.* bicuculline methiodide, kainic acid, pilocarpine, *etc.*), seizure-induced alteration of the BBB has been demonstrated (Fig. 5). Morphological and functional alterations in BBB structure can be evaluated by assessing the intravascular vs. extravascular fluorescein isothiocyanate (FITC)-albumin signal (Librizzi et al., 2012; Fig. 5B). We described a significant brain extravasation of FITC-albumin after seizure strictly depended on the time spent in seizure activity (Librizzi et al., 2012). BBB integrity was also analyzed by measuring the TJ (ZO-1) protein arrangement. BBB damage is also associated with ZO-1 down-regulation in endothelial cells (Librizzi et al., 2012; Zehendner et al., 2013). BBB permeability during electrophysiological experiments in the IWB can be monitored by evaluating changes in the extracellular K^+ concentration in the brain parenchyma (Librizzi et al., 2001; Gnatkovsky et al., 2008; Kuhlmann et al., 2009). We demonstrated that brief arterial administrations of a bolus of high K^+ solution (14.2 mM) do not change parenchymal K^+ concentration unless the BBB is impaired (Librizzi et al., 2001; Kuhlmann et al., 2009). Finally, BBB permeability can be evaluated also by testing the effect of BBB-impermeable substances. In a series of experiments, the effect of BBB-impermeable atropine (atropine-bromide) on carbachol-induced fast oscillatory activity was utilized to evaluate BBB permeability preservation (Librizzi et al., 2001).

In addition to BBB changes, the use of the IWB has been valuable to test the induction of non-constitutive adhesion factors (such as ICAM and selectins), BBB tight junction molecules (ZO-1) and pro-inflammatory molecules (IL-1beta; Librizzi et al., 2012) after the induction of seizures by several pro-epileptic agents, such as BMI, 4AP, kainic acid and pilocarpine.

3. Region-specific epileptiform patterns in the IWB

The most peculiar advantage of the use of the *in vitro* IWB for the study of ictogenesis is the ability to record *simultaneously* activities from widespreadcortical (and subcortical) systems without spatial restrictions in comparison to the *in vivo* condition. One of the most interesting finding obtained with the use of the IWB in our and other laboratories (Federico et al., 1994; Federico and MacVicar, 1996) is the demonstration that either systemic (arterial) or local intracerebral application of pro-epileptic drugs induces region-specific and to some extent system-specific interictal and seizure patterns.

Brief arterial perfusions of the $GABA_A$ receptor antagonist, BMI (50 μM for 3 min), induces ictal events in the limbic region, but generates periodic interictal spiking in the olfactory cortex (Librizzi and de Curtis, 2003). Local periodic spikes in the piriform cortex were proposed to prevent invasion of this structure by seizure activity generated in limbic structures, suggesting a "protective" effect of interictal spiking against seizure-genesis (Librizzi and de Curtis, 2003; see also de Curtis and Avanzini, 2001). BMI-induced seizure activity in limbic cortices initiate with isolated complex ISs in the CA1-EC region, that are independent and asynchronous from IS observed in the PC. Local CA1-EC potentials increased in amplitude and progressively entrain a sequence of fast rhythmic activity at 20 Hz, followed by large amplitude irregular and then bursting activity that propagates bilaterally to the hippocampal-EC system (Uva et al., 2005) and, in the absence of independent and periodic spiking in the PC, secondarily invade the olfactory region. At the end of the ictal hippocampus-EC event, periodic ISs re-establish in the PC suggesting that ictal-like activity in the PC can be recorded exclusively in the absence of an active IS condition. In fact, reinforcement of active and periodic interictal spiking induced by local injection of BMI in the PC, hinders the occurrence of secondarily ictal-like discharges in the PC after the hippocampal-EC events.

Arterial applications of the K^+ channel blocker, 4-aminopyridine (4AP; 50 μM; 4-min), induce epileptiform activities in the olfactory and limbic regions that, despite the anatomical connection between the two areas, show a completely different patterns (Fig. 6; Carriero et al., 2010; Uva et al., 2009, 2013). In the olfactory region, epileptiform events are characterized by runs of fast activity at *circa* 30 Hz followed by ictal events that originate in PC and propagate in the adjacent areas (olfactory tubercle, lateral entorhinal cortex and lateral amygdala Uva et al., 2013). This activity can evolve into seizure-like pattern, characterized by depolarization of membrane potential of PC neurons with fast activity at 60 Hz followed by large amplitude spikes. This sequence of epileptiform activity phases is independently generated in the two contralateral PCs. Intracellular recordings from layers II and III of PC demonstrated the II layers cells involvement, whereas only a subpopulation of layer III cells

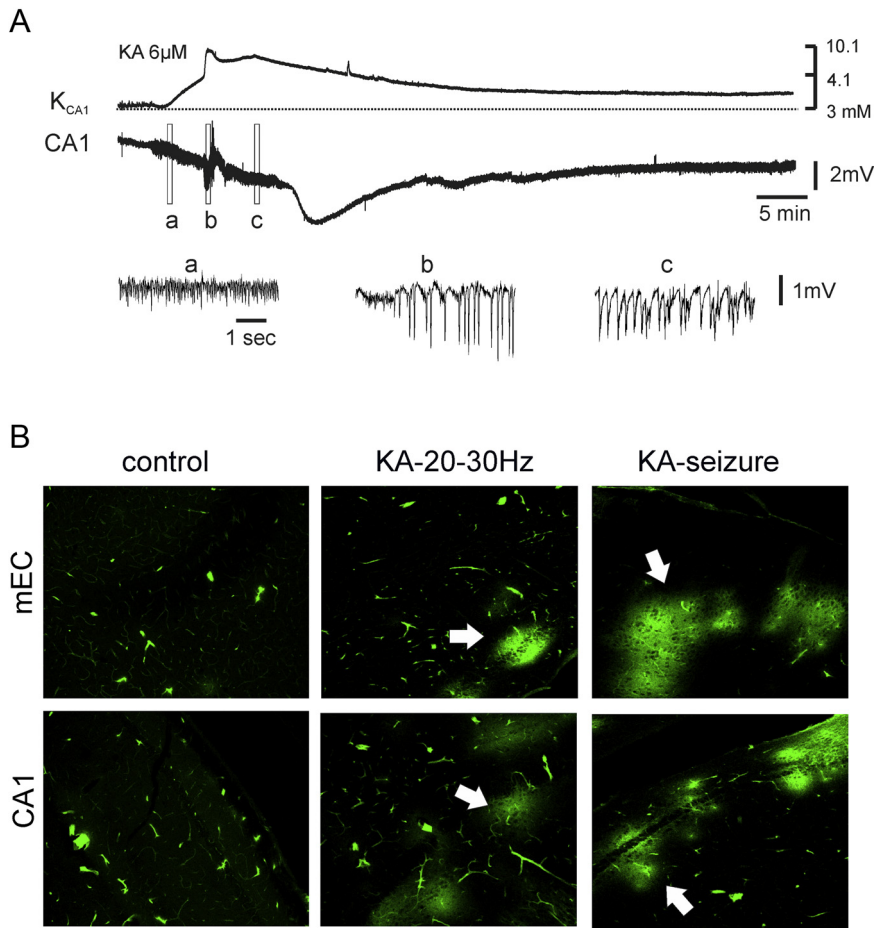

Fig. 5. (A) Simultaneous recordings of extracellular potential and changes in $[K^+]_0$ during a seizure-like event induced by arterial perfusion of KA (6 μM; 50 ml). Expended traces are illustrated in a, b, and c. The KA-induced oscillatory activity in the beta-gamma frequency range (20–30 Hz) induces a slow increase in $[K^+]_0$ followed by an abrupt and time locked rise of $[K^+]_0$ in correspondence of the occurrence of the seizure. (B) Photomicrographs of FITC–albumin immunostaining in control condition (left), during KA-induced 20–30 Hz activity (middle) and during seizure-like events (right) in the medial entorhinal cortex (mEC; upper row) and in CA1 area of the hippocampus (lower row). Areas of FITC – albumin parenchymal extravasation around vessels (white arrows) appear following KA-induced oscillatory activity (20–30 Hz) and – seizure like event.

Table 1
Advantages and limitations of the *in vitro* IWB in comparison to other experimental preparations.

	In vivo condition	*In vitro* IWB	*In vitro* slices
Preservation of connections	Unrestricted	Unrestricted	Limited by the slicing procedure
Preservation of local networks	Unrestricted	Unrestricted	Limited by the slicing procedure
Preservation of neurovascular interactions	Unrestricted but hardly accessible	Unrestricted	Not available – strongly limited by the slicing procedure
Accessibility to brain regions	Limited placement of depth recording electrodes	Virtually unrestricted	Restricted to the sliced region
Accessibility to both hemispheres	Restricted surgical access and limited electrode number	Virtually unrestricted	Unavailable
Accessibility to local networks	Restricted surgical access and limited electrode number	Virtually unrestricted	Restricted by the slicing procedure
Accessibility to multiunit activities	Spatially restricted	Virtually unrestricted	Restricted by the slicing procedure
Accessibility to single cell recording	Spatially restricted	Virtually unrestricted	Facilitated
Imaging: VSD and intrinsic signals	Limited access	Virtually unrestricted	Restricted by the slicing procedure
Calcium and cellular imaging	Limited access	Virtually unrestricted	Restricted by the slicing procedure
Systemic pharmacological manipulations	Limited by side effects of drugs and by the BBB permeability	Facilitated	Facilitated
Local pharmacological manipulations	Limited by side effects of drugs and by the limited access	Facilitated	Facilitated
Duration of recording	Virtually unrestricted	Limited to 5–7 h	Limited to 4–5 h
Study of BBB permeability	Not available	Facilitated	Not available
Study of neurovascular interactions	Possible, with spatial limitations	Virtually unrestricted	Limited by the lack of vessel integrity
Type of animal	Any	Limited to guinea pigs, prochyemis, rabbits	Any

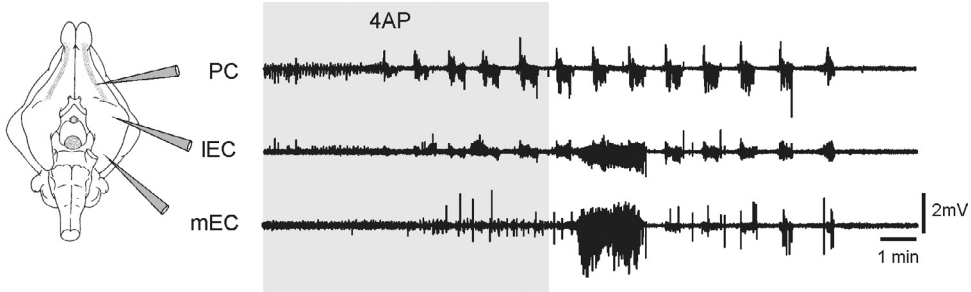

Fig. 6. On the left, a scheme of the isolated guinea-pig brain showing the position of the recording glass pipettes inserted in piriform cortex (PC), lateral entorhinal cortex (lEC) and medial entorhinal cortex (mEC). When the K^+-channel blocker 4-aminopyridine was arterially perfused in the isolated brain (50 μM; 4-min), independent seizure-like events were elicited in the olfactory and in limbic structures (compare PC/lEC and mEC traces).

discharge during the whole event. During FRs layer II cells depolarize and discharge; at SLEs onset, cells further depolarize, but firing is progressively blocked and resumes during spikes activity (Uva et al., 2013).

In the limbic cortices (*i.e.* hippocampal and mEC areas; asterisk in Fig. 6) epileptiform activity induced by 4AP starts with preictal spikes anticipating the ictal onset characterized by fast activity (20–30 Hz), followed by irregular spiking and burst activities. As in BMI-induced activity, intracellular recordings during pre-ictal spikes show that the majority of principal cells are inhibited (Uva et al., 2015) suggesting a pivotal role of inhibitory circuits ahead of a seizure. Glutamate receptors blockade with arterial application of DNQX (50 μM) and AP5 (100 μM) followed by co-perfusion with 4AP reveals recurrent interictal spikes in mEC that correlate with hyperpolarization in principal EC neurons (Uva et al., 2009). Unlike PC, seizure-like activity in the limbic system simultaneously occurs in the two hemispheres (Carriero et al., 2010).

Arterial application of kainic acid (KA; 6 μM, 50 ml) in the IWB, induced a seizure-like discharge in the limbic region. Simultaneous recordings in the PC, neocortex (NC), CA1 and/or dentate gyrus (DG), and mEC demonstrated the occurrence of seizures restricted to the hippocampus/mEC (Fig. 5A), that appear simultaneously in both hemispheres. At onset the discharge is characterized by bilateral low amplitude spike activity in the hippocampal formation and by sharp waves in mEC. An increase in spikes amplitude associated with a decrease in frequency characterizes the irregular phase and is followed by the appearance of bilateral bursts in all hippocampal areas. Interestingly, neither PC nor NC are directly involved into seizure activity and rather they show an almost complete attenuation of the electrophysiological signal amplitude, with a recover of the EEG activity within 30 min after the end of KA perfusion (unpublished observations). These experiments confirmed that specific cortical systems are able to generate seizure-like and interictal patterns that are specific and independent from neighboring areas.

4. Conclusions

The IWB is an ideal preparation to study network and cellular details of the propagation of epileptiform activity in different cortical systems. It is also an unique model to study interactions between neuronal activity and extracellular/BBB/vascular compartments in the generation of epileptiform patterns. Nevertheless, detailed experiments on the cellular mechanisms of seizure generation/progression that are feasible in less complex preparations, such as brain slices, is not allowed in the IWB preparation. Moreover, the possibility to perform effective local application of blocking drugs when the whole limbic region of the IWB is massively activated during an ictal event is limited in this close-to-*in vivo* condition.

However, the main limitation of the IWB is still the animal species constraint. The IWB has been developed in the guinea pig and in other rodents formerly identified as lagomorphs, such as the proechymis and the rabbit (Cattani A., personal communication). Several attempts to isolate *in vitro* brains of rats and mice failed, possibly because of the differences in vascular features of these animal species. It has been demonstrated that the organization of the arterial circle of Willis is, indeed, different in guinea pigs compared to mice and rats (de Curtis et al., 2012). Moreover, the lack of commercial antibodies specifically acting against guinea-pig antigens cannot be underestimated. Advantages and disadvantages of the *in vitro* IWB in comparison to other *in vitro* models and to the *in vivo* condition are summarized in Table 1. As future advancement, we plan to isolate brains from chronically epileptic guinea pigs (Arcieri et al., 2014; Carriero et al., 2012) to study network changes induced by the epileptogenic process.

Acknowledgements

This work was supported by the Italian Ministry of Health.

References

Arcieri S, Velotti R, Noe F, Carriero G, Cattalini A, Galbardi B, Gnatkovsky V, de Curtis M. Variable electrobehavioral patterns during focal nonconvulsive status epilepticus induced by unilateral intrahippocampal injection of kainic acid. Epilepsia 2014;55:1978–85.
Biella G, de Curtis M. Associative synaptic potentials in the piriform cortex of the isolated guinea-pig brain in vitro. Eur J Neurosci 1995;7:54–64.
Biella G, de Curtis M. Olfactory inputs activate the medial entorhinal cortex via the hippocampus. J Neurophysiol 2000;83:1924–31.
Biella G, Panzica F, de Curtis M. Interactions between associative synaptic potentials in the piriform cortex of the in vitro isolated guinea pig brain. Eur J Neurosci 1996;8:1350–7.
Biella G, Uva L, de Curtis M. Associative potentials in the pathway from the neocortex to the entorhinal cortex of the guinea-pig. Soc Neurosci Abstr 2000;26:703.
Biella G, Uva L, de Curtis M. Propagation of neuronal activity along the neocortical-perirhinal-entorhinal pathway in the guinea pig. J Neurosci 2002;22:9972–9.
Biella GR, Gnatkovsky V, Takashima I, Kajiwara R, Iijima T, de Curtis M. Olfactory input to the parahippocampal region of the isolated guinea pig brain reveals weak entorhinal-to-perirhinal interactions. Eur J Neurosci 2003;18:95–101.
Biella G, Spaiardi P, Toselli M, de Curtis M, Gnatkovsky V. Functional interactions within the parahippocampal region revealed by voltage-sensitive dye imaging in the isolated guinea pig brain. J Neurophysiol 2010;103:725–32.
Boido D, Jesuthasan N, de Curtis M, Uva L. Network dynamics during the progression of seizure-like events in the hippocampal-parahippocampal regions. Cereb Cortex 2014;24:163–73.
Bragin A, Engel JJ, Wilson CL, Vizentin E, Mathern GW. Electrophysiologic analysis of a chronic seizure model after unilateral hippocampal KA injection. Epilepsia 1999;40:1210–21.
Bragin A, Wilson CL, Fields T, Fried I, Engel JJ. Analysis of seizure onset on the basis of wideband EEG recordings. Epilepsia 2005;46(Suppl 5):59–63.
Carriero G, Uva L, Gnatkovsky V, de Curtis M. Distribution of the olfactory fiber input into the olfactory tubercle of the in vitro isolated guinea pig brain. J Neurophysiol 2009;101:1613–9.
Carriero G, Uva L, Gnatkovsky V, Avoli M, de Curtis M. Independent epileptiform discharge patterns in the olfactory and limbic areas of the in vitro isolated Guinea pig brain during 4-aminopyridine treatment. J Neurophysiol 2010;103:2728–36.

Carriero G, Arcieri S, Cattalini A, Corsi L, Gnatkovsky V, de Curtis M. A guinea pig model of mesial temporal lobe epilepsy following nonconvulsive status epilepticus induced by unilateral intrahippocampal injection of kainic acid. Epilepsia 2012;53:1917–27.
Cornford EM. Epilepsy and the blood brain barrier: endothelial cell responses to seizures. Adv Neurol 1999;79:845–62.
de Curtis M, Avanzini G. Interictal spikes in focal epileptogenesis. Prog Neurobiol 2001;63:541–67.
de Curtis M, Gnatkovsky V. Reevaluating the mechanisms of focal ictogenesis: the role of low-voltage fast activity. Epilepsia 2009;50:2514–25.
de Curtis M, Pare D, Llinas RR. The electrophysiology of the olfactory-hippocampal circuit in the isolated and perfused adult mammalian brain in vitro. Hippocampus 1991;1:341–54.
de Curtis M, Biella G, Forti M, Panzica F. Multifocal spontaneous epileptic activity induced by restricted bicuculline ejection in the piriform cortex of the isolated guinea pig brain. J Neurophysiol 1994;71:2463–75.
de Curtis M, Biella G, Buccellati C, Folco G. Simultaneous investigation of the neuronal and vascular compartments in the guinea pig brain isolated in vitro. Brain Res Protoc 1998a;3:221–8.
de Curtis M, Manfridi A, Biella G. Activity-dependent pH shifts and periodic recurrence of spontaneous interictal spikes in a model of focal epileptogenesis. J Neurosci 1998b;18:7543–51.
de Curtis M, Radici C, Forti M. Cellular mechanisms underlying spontaneous interictal spikes in an acute model of focal cortical epileptogenesis. Neuroscience 1999a;88:107–17.
de Curtis M, Takashima I, Iijima T. Optical imaging of cortical activity after in vitro perfusion of cerebral arteries with a voltage-sensitive dye. Brain Res 1999b;837:314–9.
de Curtis M, Librizzi L, Uva L, Gnatkovsky V. Neuronal networks in the in vitro isolated guinea pig brain. In: Ballany K, editor. Isolated central nervous system circuits. New York: Springer Science Business Media; 2012. p. 357–83 [Chapter 2012].
de Vries HE, Kuiper J, de Boer AG, Van Berkel TJ, Breimer DD. The blood–brain barrier in neuroinflammatory diseases. Pharmacol Rev 1997;49:143–55.
Dickson CT, Biella G, de Curtis M. Slow periodic events and their transition to gamma oscillations in the entorhinal cortex of the isolated Guinea pig brain. J Neurophysiol 2003;90:39–46.
Federico P, MacVicar BA. Imaging the induction and spread of seizure activity in the isolated brain of the guinea pig: the roles of GABA and glutamate receptors. J Neurophysiol 1996;76:3471–92.
Federico P, Borg SG, Salkauskus AG, Mac Vicar BA. Mapping patterns of neuronal activity and seizure propagation by imaging intrinsic optical signals in the isolated whole brain of the guinea pig preparation. Neuroscience 1994;58:461–80.
Forti M, Biella G, Caccia S, de Curtis M. Persistent excitability changes in the piriform cortex of the isolated guinea-pig brain after transient exposure to bicuculline. Eur J Neurosci 1997;9:435–51.
Gnatkovsky V, de Curtis M. Hippocampus-mediated activation of superficial and deep layer neurons in the medial entorhinal cortex of the isolated guinea pig brain. J Neurosci 2006;26:873–81.
Gnatkovsky V, Uva L, de Curtis M. Topographic distribution of direct and hippocampus-mediated entorhinal cortex activity evoked by olfactory tract stimulation. Eur J Neurosci 2004;20:1897–905.
Gnatkovsky V, Wendling F, de Curtis M. Cellular correlates of spontaneous periodic events in the medial entorhinal cortex of the in vitro isolated guinea pig brain. Eur J Neurosci 2007;26:302–11.
Gnatkovsky V, Librizzi L, Trombin F, de Curtis M. Fast activity at seizure onset is mediated by inhibitory circuits in the entorhinal cortex in vitro. Ann Neurol 2008;64:674–86.
Kuhlmann CR, Librizzi L, Closhen D, Pflanzner T, Lessmann V, Pietrzik CU, de Curtis M, Luhmann HJ. Mechanisms of C-reactive protein-induced blood–brain barrier disruption. Stroke 2009;40:1458–66.
Librizzi L, de Curtis M. Epileptiform ictal discharges are prevented by periodic interictal spiking in the olfactory cortex. Ann Neurol 2003;53:382–9.
Librizzi L, Janigro D, De Biasi S, de Curtis M. Blood brain barrier preservation in the *in vitro* isolated guinea-pig brain preparation. J Neurosci Res 2001;66:289–97.
Librizzi L, Noe F, Vezzani A, de Curtis M, Ravizza T. Seizure-induced brain-borne inflammation sustains seizure recurrence and blood–brain barrier damage. Ann Neurol 2012;72:82–90.
Marchi N, Angelov L, Masaryk T, Fazio V, Granata T, Hernandez N, Hallene K, Diglaw T, Franic L, Najm I, Janigro D. Seizure-promoting effect of blood–brain barrier disruption. Epilepsia 2007;48:732–42.
Mazzetti S, Librizzi L, Frigerio S, de Curtis M, Vitellaro-Zuccarello L. Molecular anatomy of the cerebral microvessels in the isolated guinea-pig brain. Brain Res 2004;999:81–90.
Muhlethaler M, de Curtis M, Walton K, Llinas R. The isolated and perfused brain of the guinea-pig in vitro. Eur J Neurosci 1993;5:915–26.
Pare D, deCurtis M, Llinas R. Role of the hippocampal-entorhinal loop in temporal lobe epilepsy: extra- and intracellular study in the isolated guinea pig brain in vitro. J Neurosci 1992;12:1867–81.
Pitkänen A, Schwartzkroin PA, Moshe SL. Models of seizures and epilepsy. Elsevier; 2006.
Trombin F, Gnatkovsky V, de Curtis M. Changes in action potential features during focal seizure discharges in the entorhinal cortex of the in vitro isolated guinea pig brain. J Neurophysiol 2011;106:1411–23.
Uva L, de Curtis M. Propagation pattern of entorhinal cortex subfields to the dentate gyrus in the guinea-pig: an electrophysiological study. Neuroscience 2003;122:843–51.
Uva L, de Curtis M. Polysynaptic olfactory pathway to the ipsi- and contralateral entorhinal cortex mediated via the hippocampus. Neuroscience 2005;130:249–58.
Uva L, Librizzi L, Wendling F, de Curtis M. Propagation dynamics of epileptiform activity acutely induced by bicuculline in the hippocampal-parahippocampal region of the isolated Guinea pig brain. Epilepsia 2005;46:1914–25.
Uva L, Strowbridge BW, de Curtis M. Olfactory bulb networks revealed by lateral olfactory tract stimulation in the in vitro isolated guinea-pig brain. Neuroscience 2006;142:567–77.
Uva L, Librizzi L, Marchi N, Noe F, Bongiovanni R, Vezzani A, Janigro D, de Curtis M. Acute induction of epileptiform discharges by pilocarpine in the in vitro isolated guinea-pig brain requires enhancement of blood–brain barrier permeability. Neuroscience 2008;151:303–12.
Uva L, Avoli M, de Curtis M. Synchronous GABA-receptor-dependent potentials in limbic areas of the in-vitro isolated adult guinea pig brain. Eur J Neurosci 2009;29:911–20.
Uva L, Trombin F, Carriero G, Avoli M, de Curtis M. Seizure-like discharges induced by 4-aminopyridine in the olfactory system of the in vitro isolated guinea pig brain. Epilepsia 2013;54:605–15.
Uva L, Breschi GL, Gnatkovsky V, Taverna S, de Curtis M. Synchronous inhibitory potentials precede seizure-like events in acute models of focal limbic seizures. J Neurosci 2015;35:3048–55.
Zehendner CM, Librizzi L, Hedrich J, Bauer NM, Angamo EA, de Curtis M, Luhmann HJ. Moderate hypoxia followed by reoxygenation results in blood–brain barrier breakdown via oxidative stress-dependent tight-junction protein disruption. PLoS ONE 2013;8:e82823.